

Concrete Mix Design: ACI Method

ACI (American concrete Institute) is the most common method used in North America established by ACI. Engineers deal with the latest ACI 211.1 method. ACI method of concrete mix design is based on the estimated weight of the concrete per unit volume.

Required data/information

- Target mean strength
- Coarse aggregate specification and gradation
- Fine aggregate specification and gradation
- Bulk specific gravity and adsorption of fine and coarse aggregate
- Mixing water requirements
- Relationship between strength vs. w/c-ratio
- Specific gravity of cement
- Optimum combination of aggregate to meet maximum density grading

ACI Mix Design Steps

1. workability – slump
2. Size of coarse aggregate
3. water and air content
4. w/c ratio
5. cement content
6. coarse aggregate content
7. fine aggregate content
8. moisture adjustment
9. trial mix

Example problem and solve procedure:

Problem 1: Design a concrete mix following ACI 211.1 mix design method and find the ingredients required for a trial mix of a set of cylinders. The concrete is required for an exterior column located above ground. Specified strength is 35 MPa at 28 days, with slump between 1 to 2 in. Max. aggregate size is $\frac{3}{4}$ in.

Specifications:

Cement type: Ordinary Portland cement, specific gravity = 3.15

Coarse aggregate:

Bulk sp. Gravity (OD) = 2.7

Adsorption capacity = 1.5%

unit-weight (dry-rodded) = 1600 kg/m³

Moisture content = 0.7%

Fine aggregate:

Bulk sp. Gravity (OD) = 2.65

Adsorption capacity = 1.9%

Fineness modulus = 2.7

Moisture content = 3.5%

Step 1: Choice of slump

- Required data: Types of construction

Process:

- i) Choice of unit (inch/mm) from user

```
disp('select 1 for mm unit or 2 for  
inch unit')  
choice1=input(' = ');
```

- ii) Table 5 contains slump values for different types of construction

```
disp('type 1 for building  
column/Beam/Reinforced wall, type 2  
for slab/footing/substructure  
wall')  
s=input('type of construction = ');  
if s==1  
    slump=input('slump (maximum  
100mm/4 inch) = '); %According to  
table  
else  
    slump=input('slump (maximum 75mm/3  
inch) = ');  
end
```

Table 5: Recommended slump values for various types of construction

Concrete construction	Slump, mm (in.)	
	Maximum*	Minimum
Reinforced foundation walls and footings	75 (3)	25 (1)
Plain footings, caissons, and substructure walls	75 (3)	25 (1)
Beams and reinforced walls	100 (4)	25 (1)
Building columns	100 (4)	25 (1)
Pavements and slabs	75 (3)	25 (1)
Mass concrete	75 (3)	25 (1)

*May be increased 25 mm (1 in.) for consolidation by hand methods, such as rodding and spading.
Plasticizers can safely provide higher slumps.
Adapted from ACI 211.1.

iii) Unit conversion

```
if choice1 == 2
    slump= 25*slump; % converting
    inch to mm
end
```

- Outcome: (from command window)

```
.....ACI Method for Mix Design.....

Step 1
slump choice (minimum 25 mm / 1 inch for all type of construction)
select 1 for mm unit or 2 for inch unit
= 2

type 1 for building column/Beam/Reinforced wall, type 2 for slab/footing/substructure wall
type of construction = 1
slump (maximum 100mm/4 inch) = 2
```

Step 2: Choice of maximum size of aggregate

- Required data: Maximum aggregate size

Process:

- i) Choice of unit (inch/mm) from user
- ii) Input size
- iii) Unit conversion

- Outcome:

```
Step 2
Input maximum size of aggregate
select 1 for mm unit or 2 for inch unit
= 2
maximum size of aggregate (maximum 150mm / 6 inch) = 0.75
```

Step 3: Estimating mixing water and air content

Entrained air is beneficial for freezing and thawing condition, usually used in colder countries. In Bangladesh usually non-air entrained concrete is used.

Required data: i) Slump ii) Maximum aggregate size

- Process: i) Using slump and maximum aggregate size to find the approximate water and air content from Table 4.

Table 4: Approximate mixing water in kg/m³ and target air content for different slumps and nominal maximum aggregate sizes*

Slump, mm	Water, kilograms per cubic meter of concrete, for indicated sizes of aggregate*							
	9.5 mm	12.5 mm	19 mm	25 mm	37.5 mm	50 mm**	75 mm**	150 mm**
Non-air-entrained concrete								
25 to 50	207	199	190	179	166	154	130	113
75 to 100	228	216	205	193	181	169	145	124
150 to 175	243	228	216	202	190	178	160	—
Approximate amount of entrapped air in non-air-entrained concrete, percent	3	2.5	2	1.5	1	0.5	0.3	0.2
Air-entrained concrete								
25 to 50	181	175	168	160	150	142	122	107
75 to 100	202	193	184	175	165	157	133	119
150 to 175	216	205	197	184	174	166	154	—
Recommended average total air content, percent, for level of exposure:†								
Mild exposure	4.5	4.0	3.5	3.0	2.5	2.0	1.5	1.0
Moderate exposure	6.0	5.5	5.0	4.5	4.5	4.0	3.5	3.0
Severe exposure	7.5	7.0	6.0	6.0	5.5	5.0	4.5	4.0

%Following values are from table

```

mixwater1= [ 207 199 190 179 166 154 130 113];
mixwater2= [ 228 216 205 193 181 169 145 124];
mixwater3= [ 181 175 168 160 150 142 122 107];
mixwater4= [202 193 184 175 165 157 133 119];
air1= [3 2.5 2 1.5 1 .5 .3 .2];
air2= 4.5:-0.5:1;
air3= [6 5.5 5 4.5 4.5 4 3.5 3];
air4= [7.5 7 6 6 5.5 5 4.5 4];
agg_size= [9.5 12.5 19 25 37.5 50 75 150];

```

linear interpolation is used here to get required air percentage % and required mixing water in kg/m³ from table

```

%for non air-entrained
if choice3==1
    required_air= interp1(agg_size,air1,agg,'linear');
    if (slump>=25 && slump <= 50 )
        required_water= interp1(agg_size,mixwater1,agg,'linear');
    elseif (slump>50 && slump <= 100 )
        required_water= interp1(agg_size,mixwater2,agg,'linear');
    end
end

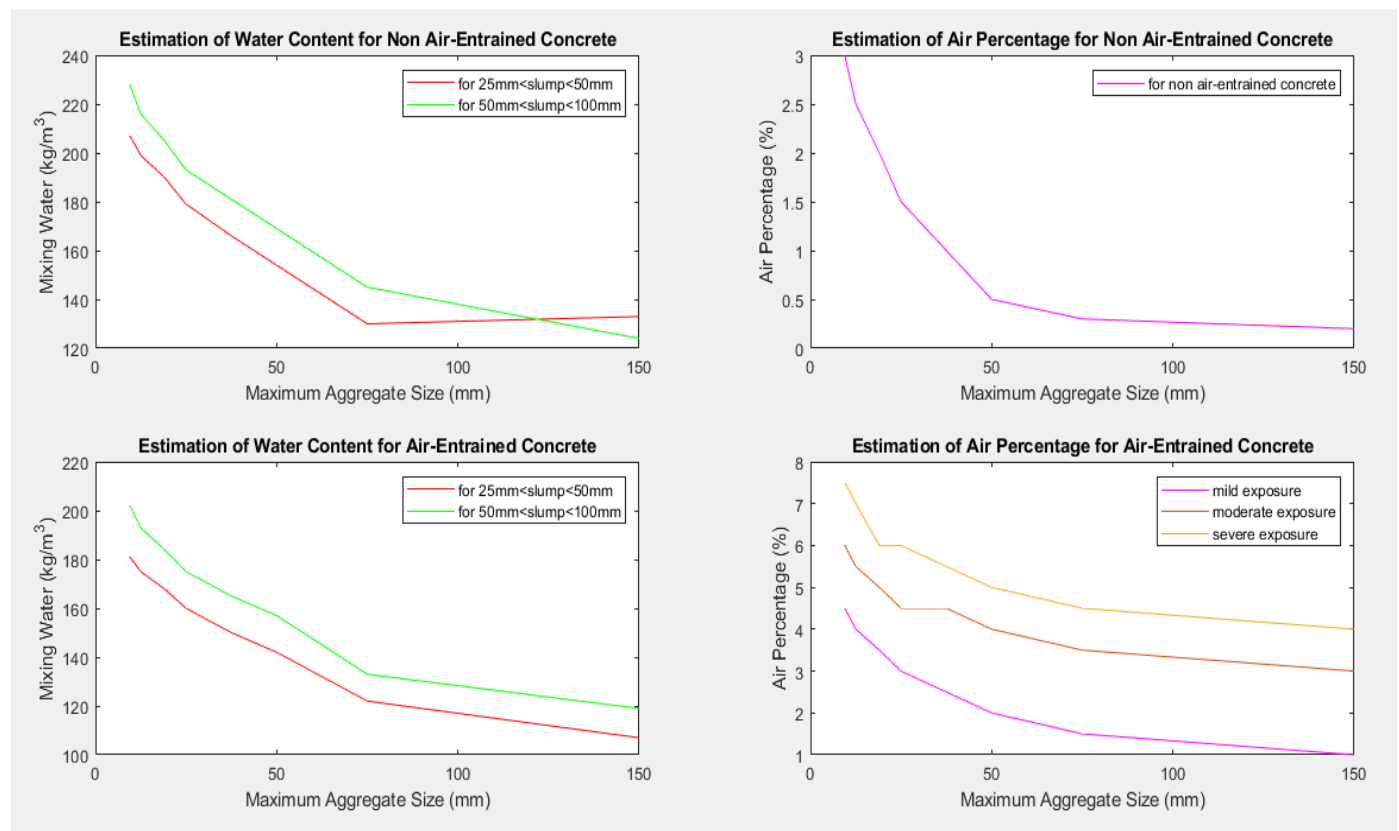
```

```

%for air-entrained
if choice3==2
    %level of exposure
    disp('select level of exposure')
    disp('type 1 for mild exposure, 2 for moderate ,3 for severe')
    choice4=input('=');
    if choice4==1 %mild exposure
        required_air= interp1(agg_size,air2,agg,'linear');
    elseif choice4==2 %moderate exposure
        required_air= interp1(agg_size,air3,agg,'linear');
    elseif choice4==3 %severe exposure
        required_air= interp1(agg_size,air4,agg,'linear');
    end
if (slump>=25 && slump <= 50 )
    required_water= interp1(agg_size,mixwater3,agg,'linear');
elseif (slump>50 && slump <= 100 )
    required_water= interp1(agg_size,mixwater4,agg,'linear');
end
end
end

```

- Outcome: Approximate water and air content



Graph generated from given table using 'subplot'

Step 3
 choose type of concrete: 1 for non air-entrained, 2 for air-entrained
 =2
 select level of exposure
 type 1 for mild exposure, 2 for moderate ,3 for severe
 =2
 Step 3 results:

	Amount in kg/m ³
Required Water	168.27
Estimated Air	5.0192

For air entrained concrete

Step 3
 choose type of concrete: 1 for non air-entrained, 2 for air-entrained
 =1
 Step 3 results:

	Amount in kg/m ³
Required Water	190.35
Estimated Air	2.0192

For non-air entrained concrete

Step 4: Selection of w/c ratio

Required data: i) Required average strength, f_{cr} ' ii) Air condition

Process: i) Overdesign necessary to meet strength requirements

Table 1 (b): Required average compressive strength when data are not available to establish a standard deviation

Specified compressive strength, f'_c , MPa	Required average compressive strength, f'_{cr} , MPa
Less than 21	$f'_c + 7.0$
21 to 35	$f'_c + 8.5$
Over 35	$1.10f'_c + 5.0$

Adapted from ACI 318.

```
% Overdesign necessary to meet strength requirements
if (strength<=21)
    Required_average_strength=strength+7;
elseif (strength>21 && strength<=35)
    Required_average_strength=strength+8.5;
elseif (strength>35)
    Required_average_strength=1.1*strength+5;
end
```

ii) Using compressive strength and air condition to find the water/cement ration from Table 2.

Table 2: Relationship between water to cementitious material ratio and compressive strength of concrete.

Compressive strength at 28 days, MPa	Water-cementitious materials ratio by mass	
	Non-air-entrained concrete	Air-entrained concrete
45	0.38	0.30
40	0.42	0.34
35	0.47	0.39
30	0.54	0.45
25	0.61	0.52
20	0.69	0.60
15	0.79	0.70

Strength is based on cylinders moist-cured 28 days in accordance with ASTM C 31 (AASHTO T 23). Relationship assumes nominal maximum size aggregate of about 19 to 25 mm.
Adapted from ACI 211.1 and ACI 211.3.

```
ST=15:5:45;
WC1= [.79 .69 .61 .54 .47 .42 .38];
WC2= [.7 .6 .52 .45 .39 .34 .3];

if choice3==1 %for non air-entrained concrete
wc_ratio=interp1(ST,WC1,Required_average_strength,
'linear');
elseif choice3==2 %for air-entrained concrete

wc_ratio=interp1(ST,WC2,Required_average_strength,
'linear');
end
```

• Outcome: water/cement ratio

Step 4

Input Target Mean Strength:

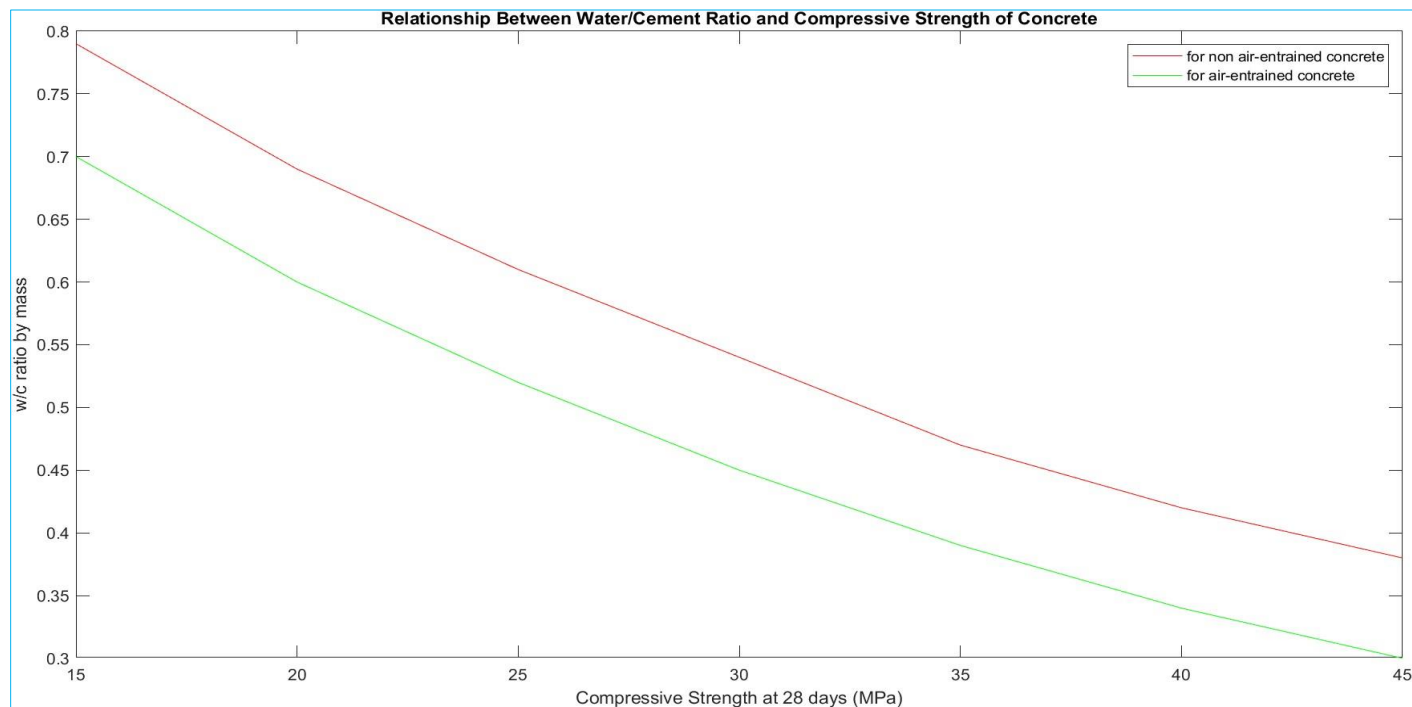
```
select unit:input 1 for MPa, 2 for Psi
= 1
target mean strength (maximum 36 MPa) =35
```

Step 4 Results:

	Amounts
Compressive Strength (MPa)	43.5
Estimated W/C Ratio	0.392

35

Given data



```
% graph generating
figure
plot(ST,WC1,'r',ST,WC2,'g')
xlabel('Compressive Strength at 28 days (MPa)'),ylabel('w/c ratio by mass')
legend('for non air-entrained concrete','for air-entrained concrete')
title('Relationship Between Water/Cement Ratio and Compressive Strength of Concrete')
```

Step 5: Determination of cement content

- Required data: i) w/c ratio: from Step 4 ii) water content: from Step 3
- Process: i) $\text{Cement content} = \text{free-water content} / \text{wc-ratio}$
- Output: ▪ Cement content in kg/m³

Current Fold	Step 5	
	step 5 Result:	
Workspace		Amount in kg/m ³
	Cement Content	485.58

Step 6: Estimating coarse aggregate content

- Required data: i) Maximum aggregate size
 - ii) Unit weight of coarse aggregate
 - iii) FM of fine aggregate
- Process:
 - i) Using FM of fine aggregate and maximum aggregate size to find the per unit volume of dry-rodded (OD) coarse aggregate from Table 3.

According to aggregate size row index of matrix was allocated here.

Table 3: Bulk volume of coarse aggregate per unit volume of concrete

Nominal maximum size of aggregate, mm (in.)	Bulk volume of dry-rodded coarse aggregate per unit volume of concrete for different fineness moduli of fine aggregate*			
	2.40	2.60	2.80	3.00
9.5 (3/8)	0.50	0.48	0.46	0.44
12.5 (1/2)	0.59	0.57	0.55	0.53
19 (3/4)	0.66	0.64	0.62	0.60
25 (1)	0.71	0.69	0.67	0.65
37.5 (1 1/2)	0.75	0.73	0.71	0.69
50 (2)	0.78	0.76	0.74	0.72
75 (3)	0.82	0.80	0.78	0.76
150 (6)	0.87	0.85	0.83	0.81

*Bulk volumes are based on aggregates in a dry-rodded condition as described in ASTM C 29 (AASHTO T 19). Adapted from ACI 211.1.

Creating matrix:
 fm= [2.4 2.6 2.8 3];
 vol_CA=[.5 .48 .46 .44;
 .59 .57 .55 .53;
 .66 .64 .62 .6;
 .71 .69 .67 .65;
 .75 .73 .71 .69;
 .78 .76 .74 .72;
 .82 .8 .78 .76;
 .87 .85 .83 .81];

```
if agg<= 9.5
    i=1; %row index
elseif (agg> 9.5 && agg<=12.5)
    i=2;
elseif (agg> 12.5 &&
agg<=19)
    i=3;
```

```
elseif (agg> 19 && agg<=25)
    i=4;
elseif (agg>25 && agg<=37.5)
    i=5;
elseif (agg> 37.5 && agg<=50)
    i=6;
```

```
elseif (agg>50 && agg<=75)
    i=7;
elseif (agg> 75 && agg<=150)
    i=8;
end
volm_CA=vol_CA(i,:);
```

ii) **OD weight of CA = OD rodded volume * bulk unit weight**

Outcome: ▪ Weight of coarse aggregate

Editor

Step 6

Unit Weight of coarse aggregate in kg/m³=1600

Input FM of Fine Aggregate (range 2.4 to 3.0)

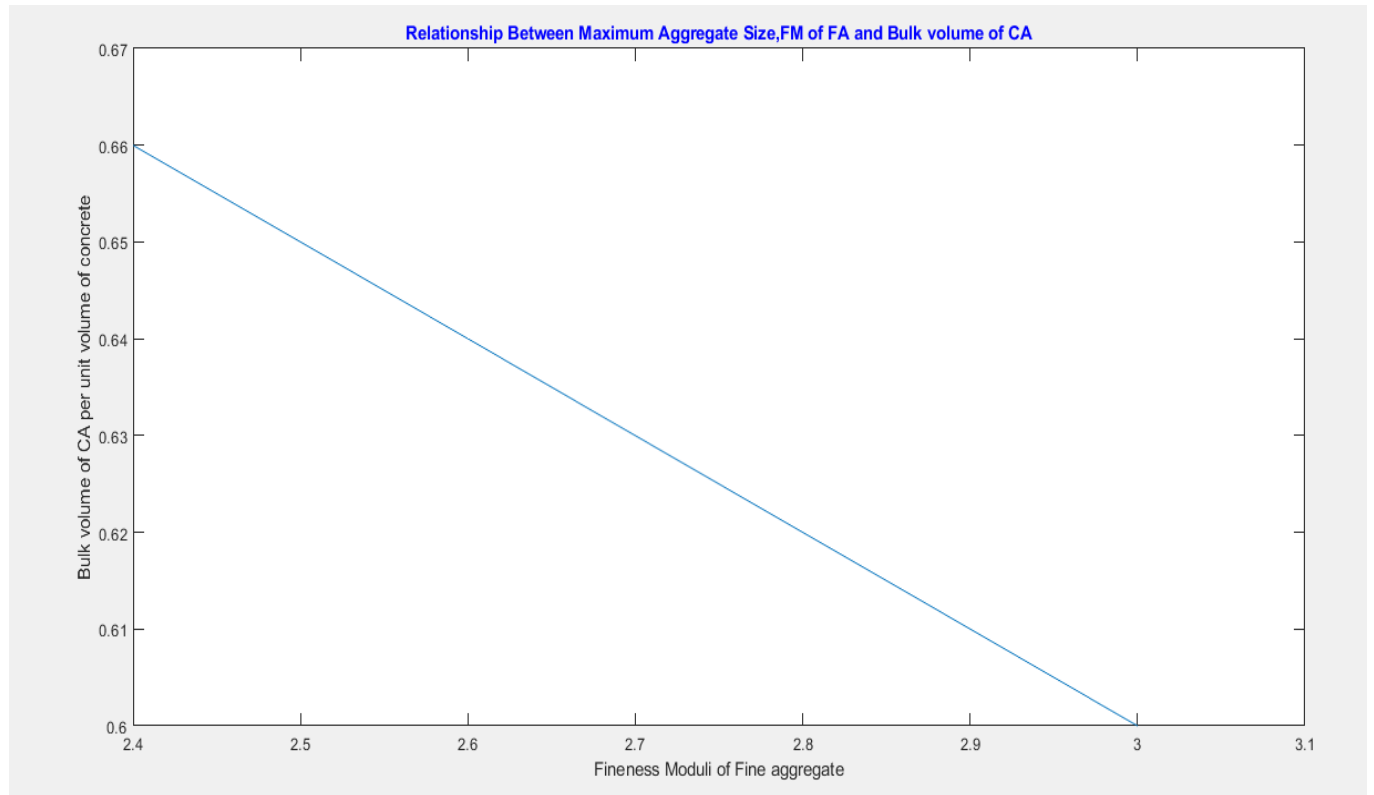
Fineness Modulus =2.7

step 6 result:

Amount in kg/m³

OD Weight of CA 1008

Given data



Step 7: Estimating fine aggregate content

Required data: i) Water content ii) Cement content iii) Coarse aggregate content

Weight basis and **Volume basis** methods are used here.

Process: i) **Weight basis**: - Using air condition and maximum aggregate size to find the estimated weight of the fresh concrete per unit volume from Table 6.

Table 6: First estimate of density (unit weight) of fresh concrete

Nominal maximum size of aggregate, mm	First estimate of concrete unit mass, kg/m ³ *	
	Non-air-entrained concrete	Air-entrained concrete
9.5	2280	2200
12.5	2310	2230
19	2345	2275
25	2380	2290
37.5	2410	2350
50	2445	2345
75	2490	2405
150	2530	2435

```

%weight basis
if choice6==1
%Table 6: First estimate of density (unit weight) of fresh concrete
concrete1=[2280 2310 2345 2380 2410 2445 2490 2530];
concrete2=[220 2230 2275 2290 2350 2345 2405 2435];
if choice3==1 %for non air-entrained concrete
concrete_density=interp1(agg_size,concrete1,agg,"linear");
elseif choice3==2 %for air-entrained concrete
concrete_density=interp1(agg_size,concrete2,agg,"linear");
end

```

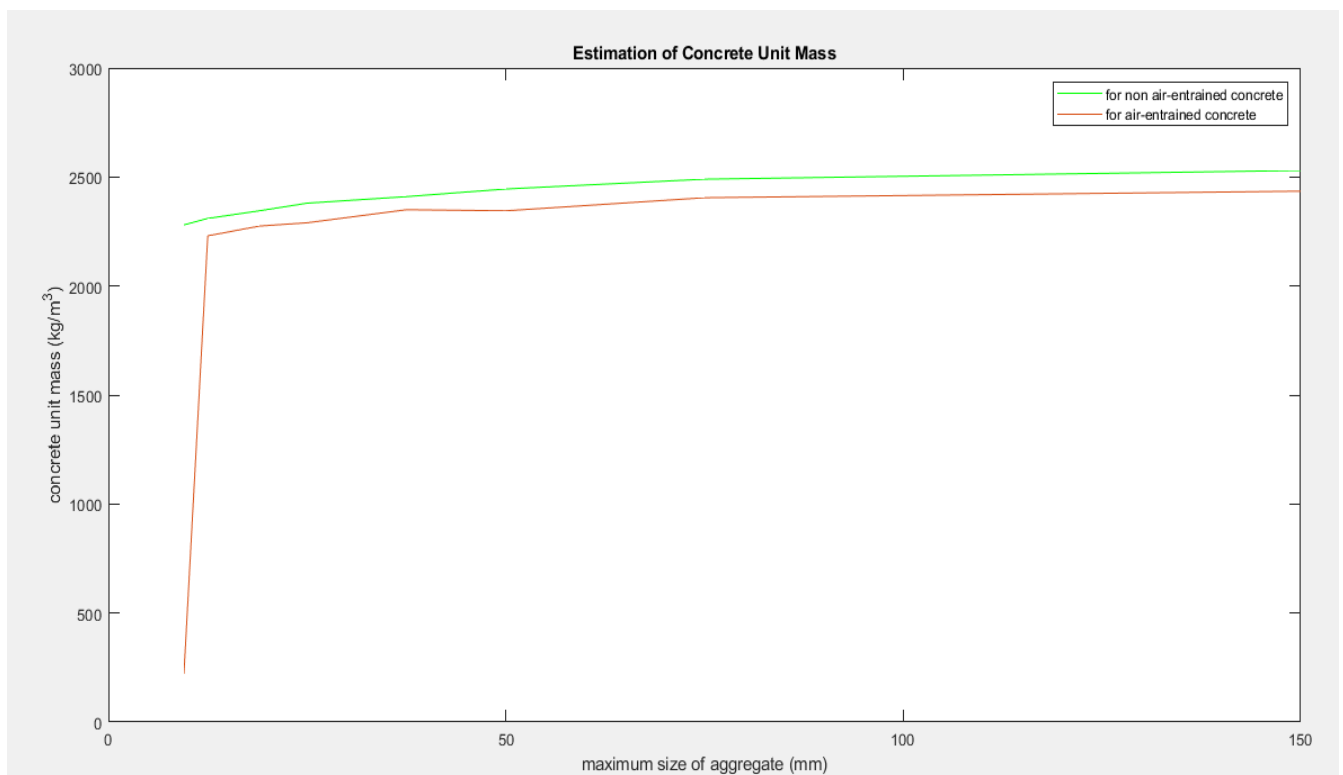
ii) FA mass(SSD) = concrete density – water mass – cement mass – CA mass(SSD)

```

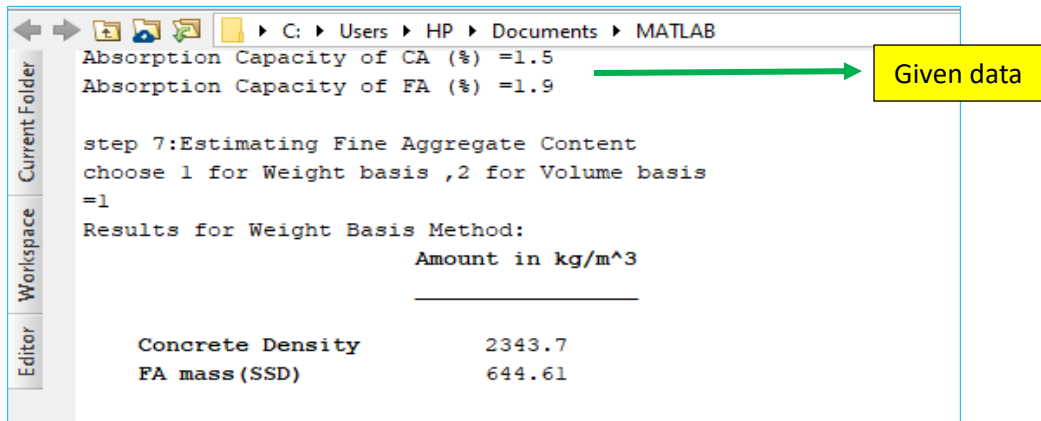
%SSD weight of CA = OD weight of CA * (1 + absorption capacity (%)/100)
SSD_CA= OD_weightCA*(1+(AC_CA/100));
%FA mass(SSD) = concrete density - water mass - cement mass - CA mass (SSD)
SSD_FA= concrete_density-required_water-cement-SSD_CA;
OD_weightFA= SSD_FA/(1+(AC_FA/100));

```

- Outcome: [Graph for Table 6:](#)



- Weight of fine aggregate



A screenshot of a MATLAB script window. The script defines variables for Absorption Capacity of CA (%) = 1.5 and Absorption Capacity of FA (%) = 1.9. It then prompts the user to choose a basis (1 for Weight, 2 for Volume) and shows the results for the Weight Basis Method. A yellow box labeled 'Given data' has an arrow pointing to the input values.

```

C:\Users\HP\Documents\MATLAB
Absorption Capacity of CA (%) =1.5
Absorption Capacity of FA (%) =1.9

step 7:Estimating Fine Aggregate Content
choose 1 for Weight basis ,2 for Volume basis
=1
Results for Weight Basis Method:
Amount in kg/m^3
Concrete Density      2343.7
FA mass(SSD)          644.61
  
```

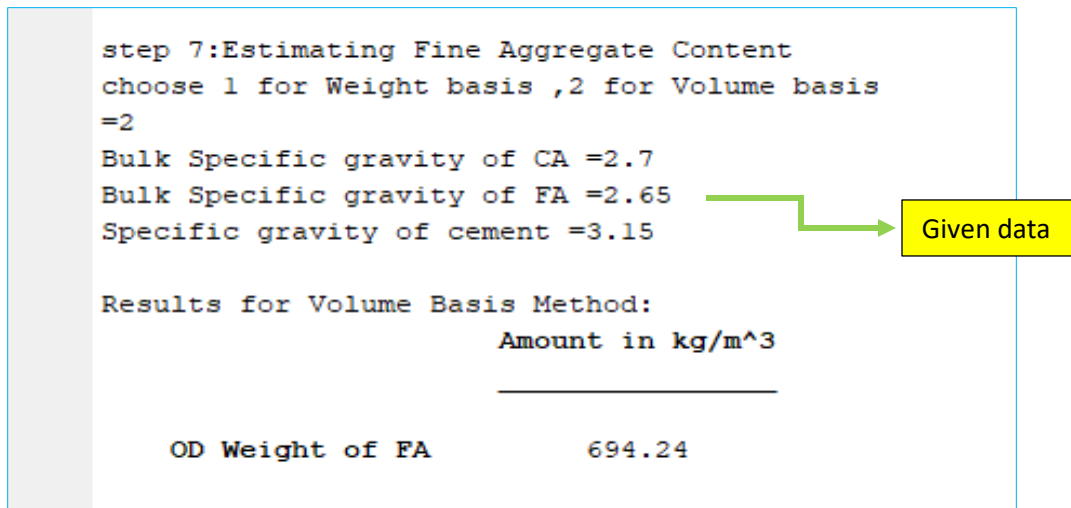
Process: ii) **Volume basis: (more accurate method)**

$$\text{Volume of Water} + \text{Cement} + \text{FA} + \text{CA} + \text{Air} = \text{Volume of concrete}$$

```

%volume =weight/(sp. gravity * densitywater)
if choice6==2
BSG_CA =input ('Bulk Specific gravity of CA =');
BSG_FA =input ('Bulk Specific gravity of FA =');
SG_cem =input ('Specific gravity of cement =');
volm_FA= 1-required_air/100-(OD_weightCA/(BSG_CA*1000)) -(required_water/1000) -
(cement/(SG_cem*1000));
OD_weightFA=volm_FA*BSG_FA*1000;
  
```

- Outcome: ▪ Weight of fine aggregate



A screenshot of a MATLAB script window showing the Volume Basis Method. It defines variables for Bulk Specific gravity of CA = 2.7, Bulk Specific gravity of FA = 2.65, and Specific gravity of cement = 3.15. It prompts the user to choose a basis (1 for Weight, 2 for Volume) and shows the results for the Volume Basis Method. A yellow box labeled 'Given data' has an arrow pointing to the input values.

```

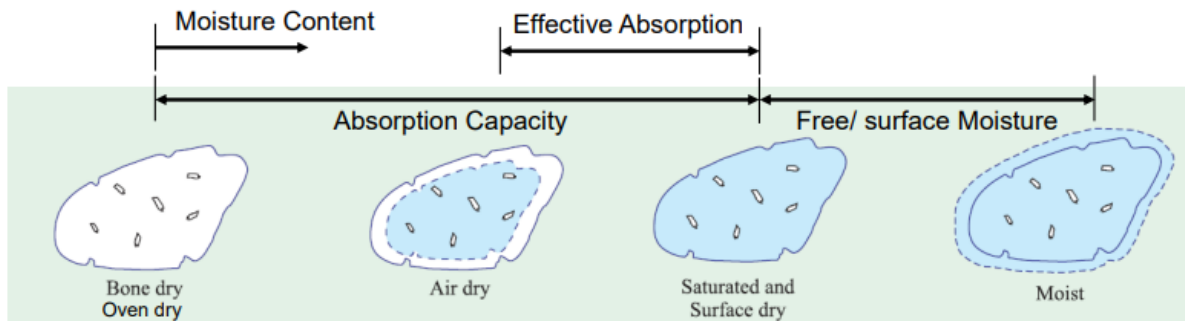
step 7:Estimating Fine Aggregate Content
choose 1 for Weight basis ,2 for Volume basis
=2
Bulk Specific gravity of CA =2.7
Bulk Specific gravity of FA =2.65
Specific gravity of cement =3.15

Results for Volume Basis Method:
Amount in kg/m^3
OD Weight of FA      694.24
  
```

Step 8: Adjustment for aggregate moisture

If aggregate contain free moisture that should be deducted from the total water content. If aggregate is at air dry condition mixing water content amount should be increased to take the aggregate to SSD condition.

Different Stages of Absorption



$$\text{Moisture Content (MC)} = \frac{\text{weight} - OD}{OD} * 100 (\%)$$

$$\text{Absorption Capacity (AC)} = \frac{SSD - OD}{OD} * 100 (\%)$$

$$\text{Surface Moisture} = MC - AC$$

Code:

```
%Water Adjustment :  $\Delta W_{water} = WOD * Surf. moist.$   
adjusted_water= required_water -(OD_weightCA*((MC_CA-AC_CA)/100))-(OD_weightFA*((MC_FA-AC_FA)/100));  
free_water_content=adjusted_water;  
fine_agg_content=field_FA;  
coarse_agg_content=field_CA;  
end %end of ACI Method
```

Summary of mix design:

Creating table to show the amounts of ingredients.

```
lm  X  +
%summary: For 1 cubic meter of compacted fresh concrete
densityfinal=cement+free_water_content+fine_agg_content+coarse_agg_content;
cement_bags=ceil(cement/50);
fprintf('\n')
disp('Mix Design Results: For 1 cubic meter of Compacted Fresh Concrete')
ve={'Amounts', 'Unit'};
row={'Cement', 'Water', 'Fine Aggregate', 'Coarse Aggregate', 'Density of Concrete'};
var1=[cement_bags,free_water_content,fine_agg_content,coarse_agg_content,densityfinal];
var2=(['bags','kg','kg','kg','kg/m^3']);
t6=table(var1,var2,'VariableNames',ve,'RowNames',row);
disp(t6)
fprintf('Mix Ratio (by Weight)= 1:%.2f :%.2f \n',fine_agg_content/cement,coarse_agg_content/cement)
```

1 bag of cement= 50 kg
Ceil is used for upper rounding.

Outcome: Result For 1 cubic meter of compacted fresh concrete (ACI)

```
step 8

Input Moisture Contents (%)
moisture content of CA(%) =.7
moisture content of FA (%) =3.5

Mix Design Results: For 1 cubic meter of Compacted Fresh Concrete
```

	Amounts	Unit
Cement	10	"bags"
Water	188.29	"kg"
Fine Aggregate	654.73	"kg"
Coarse Aggregate	1015.1	"kg"
Density of Concrete	2343.7	"kg/m^3"

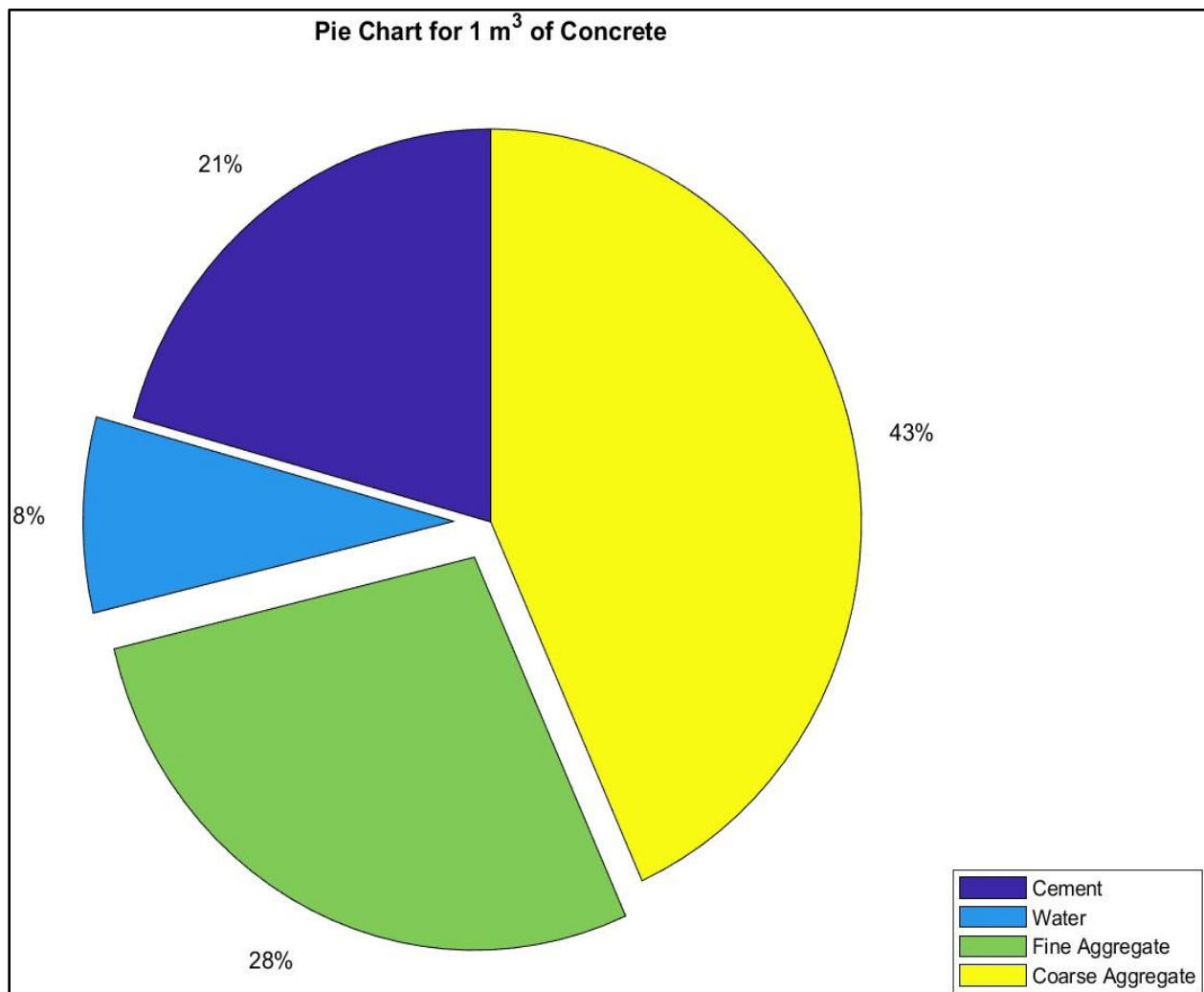
```
fx Mix Ratio (by Weight)= 1:1.35 :2.09
```

Mix ratio

Pie chart:

```
%pie chart
labels={'Cement','Water','Fine Aggregate','Coarse Aggregate'};
X=[cement,free_water_content,fine_agg_content,coarse_agg_content];
explode = [0 1 1 0];
figure
pie(X,explode)
legend(labels,'location','southeastoutside')
title('Pie Chart for 1 m3 of Concrete')
```

Output: (ACI method)



Cost estimation:

```
%cost estimation
fprintf('\nTo Calculate Cost of Concrete press 1\n')
Choice10=input('=');
if Choice10 == 1
    cement_cost=input('Per Bag Cement cost (in BDT)=');
    fa_cost=input('Per m^3 Fine Aggregate cost (in BDT)=');
    ca_cost=input('Per m^3 Coarse Aggregate cost (in BDT)=');
    fprintf('\n')
    disp('Cost Estimation: For 1 cubic meter of fresh concrete')
    ve_c={'Amounts', 'Per unit cost','Total cost'};
    row_c={'Cement Bag','Fine Aggregate','Coarse Aggregate'};
    var_c1=[cement_bags,fine_agg_content,coarse_agg_content]';
    var_c2=[cement_cost,fa_cost,ca_cost]';
    var_c3=var_c1.*var_c2;
    t7=table(var_c1,var_c2,var_c3,'VariableNames',ve_c,'RowNames',row_c);
    disp(t7)
    fprintf('Total cost of concrete = %.2f Tk.\n',sum(var_c3))
end
```

Output:

To Calculate Cost of Concrete press 1
=1
Per Bag Cement cost (in BDT)=500
Per m³ Fine Aggregate cost (in BDT)=30
Per m³ Coarse Aggregate cost (in BDT)=100

Cost Estimation: For 1 cubic meter of fresh concrete

	Amounts	Per unit cost	Total cost
Cement Bag	10	500	5000
Fine Aggregate	654.73	30	19642
Coarse Aggregate	1015.1	100	1.0151e+05

Total cost of concrete = 126147.57 Tk.

Inputs from user

Total cost

Adjustment for trial batch:

- Determination of the volume of specimens to be casted
- Applying loss factor/shrinkage factor
- Determination of mass of each material

end

%Step 10:Adjustment for trial batch

if choice7==1

disp('select number of specimen to be casted')

n=input('no of specimens=');

fprintf('\n')

disp('select type of specimen to be casted')

disp('select 1 for cylinder,2 for rectangular,3 for square block')

choice8=input('specimen type=');

fprintf('\n')

disp('unit of dimensions:1 for meter,2 for feet')

choice9=input('unit choice=');

fprintf('\n')

height=input('height=');

3 types of specimens are considered.

- i) Cylinder
- ii) Rectangular block
- iii) Square block

if choice8==1

radius=input('radius=');

trial_volm=1.5*n*height*pi*radius^2;

Volume of cylinder

elseif choice8==2

length1=input('length=');

width=input('width=');

trial_volm=1.5*n*height*width*length1;

Volume of rectangle

elseif choice8==3

trial_volm=1.5*n*height^3;

Volume of square

end

if choice9==2

trial_volm=trial_volm*(.3048)^3;%converting to m^3

end

fprintf('\nVolume of Trial Mix = %f m^3 \n',trial_volm)

%mass for trial volume

fprintf('\nMix Design Results: For trial mix\n')

ve1={'Amounts for 1 m^3','Amounts for Trial Mix','Unit'};

var3=trial_volm.*var1;

var3(1)=ceil(var3(1));

Ceil is used for upper rounding.

t8=table(var1,var3,var2,'VariableNames',ve1,'RowNames',row);

disp(t8)

end

Output from command window:

```
FILE          VARIABLE          CODE
C:\Users\HP\Documents\MATLAB

Do You Want to Calculate Amount for Trial Mix? press 1 ,else press 2
=1
select number of specimen to be casted
no of specimens=6
select type of specimen to be casted
select 1 for cylinder,2 for rectangular,3 for square block
specimen type=1
unit of dimensions:1 for meter,2 for feet
unit choice=1
height=0.5
radius=0.3
Volume of Trial Mix = 1.272345 m^3
Mix Design Results: For trial mix
```

	Amounts for 1 m^3	Amounts for Trial Mix	Unit
Cement	10	13	"bags"
Water	188.29	239.57	"kg"
Fine Aggregate	654.73	833.05	"kg"
Coarse Aggregate	1015.1	1291.5	"kg"
Density of Concrete	2343.7	2981.9	"kg/m^3"

Sieve Analysis of Fine Aggregate:



Example Problem 1: Sieve Analysis of Fine Aggregate

Sieve Size (mm)	Mass Retained (g)	Individual Retained (%)	Cumulative Retained (%)	% Finer/ Total Passing (%)
4.75 (No. 4)	3			
2.36 (No. 8)	29			
1.18 (No. 16)	65			
0.60 (No. 30)	165			
0.30 (No. 50)	131			
0.15 (No. 100)	97			
0.075 (No. 200)	10			

Code:

```
%sieve ananlysis of Fine Aggregate
sieve_no=[4 8 16 30 50 100 200]';
sieve_size=[4.75 2.36 1.18 .6 .3 .15 .075]';

if choice1==2 %sieve ananlysis of Coarse Aggregate
    sieve_size=[25 19 12.5 9.5 4.75 2.36 ]';
end
j=length(sieve_size);
mass=zeros(j,1);
for i=1:j
    fprintf('mass retained for sieve size %.2f mm',sieve_size(i))
    mass(i)=input('=');
end
total=sum(mass);
individual_retained=zeros(j,1);
for i=1:j
    individual_retained(i)=round((mass(i)*100)/total);
end
cumulative_retained=cumsum(individual_retained);%cumulative sum
finer=100-cumulative_retained;
if choice1==1
    FM1=sum(cumulative_retained(1:j-1))/100;
```

'For' loop used here to get inputs from user

FM is obtained by adding the cumulative % retained on specified sieve (ASTM standard sieves) and dividing the sum by 100.

$$FM = \sum \text{Cumulative \% retained on standard sieves} / 100$$

```

%table create
ve2={'Sieve No.','Sieve Size(mm)','Mass Retained (g)','Individual Retained(%)','Cumulative Retained(%)','% Finer'};
t5=table(sieve_no,sieve_size,mass,individual_retained,cumulative_retained,finer,'VariableNames',ve2);
disp(t5)
fprintf('Fineness modulus is %.2f \n',FM1)

```

```

%semilog graph generating
figure
semilogx(sieve_size,finer,'-o',MarkerFaceColor='r')
grid on
title('Gradation Curve','FontSize',15)
xlabel('Sieve opening (mm)'),ylabel('%Finer')
legend('for fine aggregate',Location='southeast')
end

```

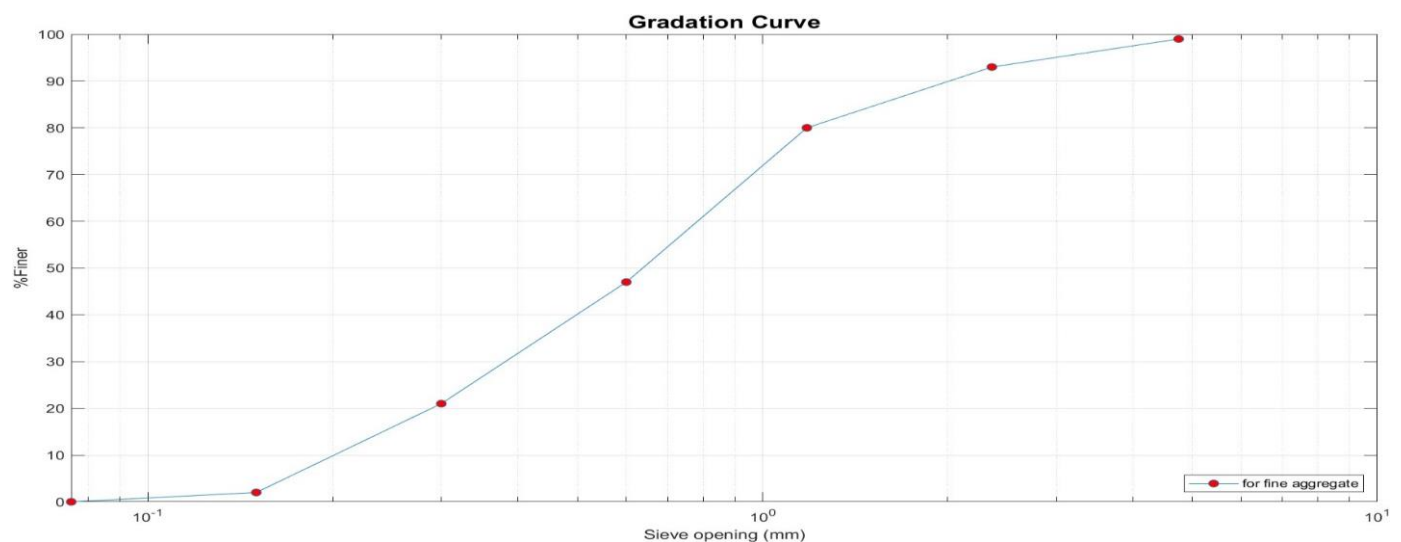
Gradation curve

```

%Coefficient of Uniformity, Cu and Coefficient of Curvature (Cc or Cz)
%Cu=D60/D10, Cz=(D30)^2/D10*D60,
D60=interp1(finier,sieve_size,60,"linear");
D10=interp1(finier,sieve_size,10,"linear");
D30=interp1(finier,sieve_size,30,"linear");
Cu=D60/D10;
Cz=(D30)^2/(D10*D60);
fprintf('Coefficient of Uniformity,Cu = %.2f \n',Cu)
fprintf('Coefficient of Curvature,Cz = %.2f \n',Cz)

```

Output:



Command window:

Current F
Workspace
Editor

```

press 1 for sieve analysis of Fine Aggregate, 2 for Coarse Aggregate
=1
mass retained for sieve size 4.75 mm=3
mass retained for sieve size 2.36 mm=29
mass retained for sieve size 1.18 mm=65
mass retained for sieve size 0.60 mm=165
mass retained for sieve size 0.30 mm=131
mass retained for sieve size 0.15 mm=97
mass retained for sieve size 0.07 mm=10
      
```

Input of values according to question

Sieve No.	Sieve Size(mm)	Mass Retained (g)	Individual Retained(%)	Cumulative Retained(%)	% Finer
4	4.75	3	1	1	99
8	2.36	29	6	7	93
16	1.18	65	13	20	80
30	0.6	165	33	53	47
50	0.3	131	26	79	21
100	0.15	97	19	98	2
200	0.075	10	2	100	0

Fineness modulus is 2.58
 Coefficient of Uniformity, $C_u = 3.89$
 Coefficient of Curvature, $C_z = 0.92$

Results


```

End of Session
.....Thank You.....
fx >>
      
```

Sieve Analysis of Coarse Aggregate:



Example Problem 2: Sieve Analysis of Coarse Aggregate

Draw gradation curve and Determine the FM of aggregates from the following particle distribution.

Sieve Size (mm)	Mass Retained (g)
25.0 (1 in.)	0
19.0 (3/4 in.)	405
12.5 (1/2 in.)	2850
9.5 (3/8 in.)	2435
4.75 (No. 4)	2030
2.36 (No. 8)	375

Command window:

End of session
press 1 for sieve analysis of Fine Aggregate, 2 for Coarse Aggregate

=2

mass retained for sieve size 25.00 mm=0
mass retained for sieve size 19.00 mm=405
mass retained for sieve size 12.50 mm=2850
mass retained for sieve size 9.50 mm=2435
mass retained for sieve size 4.75 mm=2030
mass retained for sieve size 2.36 mm=375

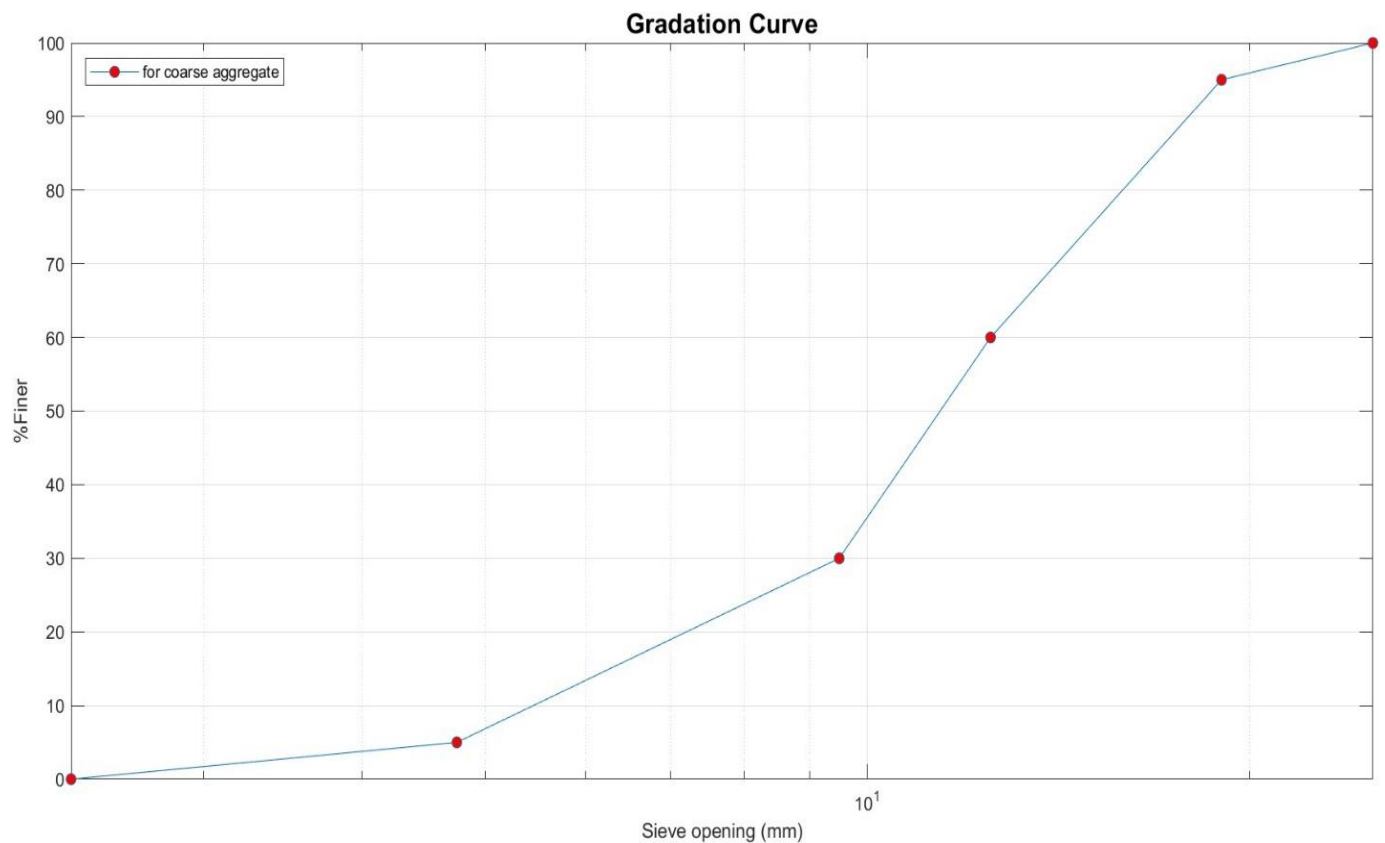
Input of values according to question

Sieve Size(mm)	Mass Retained (g)	Individual Retained(%)	Cumulative Retained(%)	% Finer
25	0	0	0	100
19	405	5	5	95
12.5	2850	35	40	60
9.5	2435	30	70	30
4.75	2030	25	95	5
2.36	375	5	100	0

Fineness modulus is 6.70
Coefficient of Uniformity, $C_u = 2.19$
Coefficient of Curvature, $C_z = 1.27$

End of Session
.....Thank You.....

fx >> |



Result For British Method:

Mix Design Results: For 1 cubic meter of Compacted Fresh Concrete

	Amounts	Unit
Cement	11	"bags"
Water	230	"kg"
Fine Aggregate	507.3	"kg"
Coarse Aggregate	1085.6	"kg"
Density of Concrete	2372.9	"kg/m ³ "

Mix Ratio (by Weight)= 1:0.92 :1.97

